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Predictive Insights Beyond Boundaries: Integrating Clinical Data for Hepatitis C Forecasting Using Machine Learning and Deep Learning

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***Abstract*—A person's well-being and wellness are seriously hampered by the chronic diseases' lasting implications and recurrent signs. A startling 58 million people worldwide suffer from an incurable chronic form of hepatitis C, disease is brought about by a virus called hepatitis C (HCV). Additionally, it results in 290,000 deaths every year from cirrhosis and liver cancer, both of which linked to HCV. The goal of this venture is to reliably identify and anticipate hepatitis C utilizing Machine Learning (ML) and deep Learning (DL) techniques. It does this by utilizing ML and DL algorithms' capacity to scan big datasets and spot patterns and associations that help with diagnosis. Predictive models were created utilizing blood values from 615 patients, including both hepatitis patients and healthy blood donors, using a variety of approaches comprising SVM, Logistic regression (LR), KNN, LSTM and others. The dataset underwent extensive preprocessing, which included managing missing values, scaling, and diminished dimensionality using Principal Component Analysis (PCA). After a variety of ML along with the SVM classifier showed the best result, giving an accuracy concerning 98.33%, recall concerning 94.17%, precision concerning 93.99%, and F1-score concerning 94.16%.**

[1] ***Keywords—Hepatitis C Virus, Machine Learning, Deep Learning, PCA, SVM.***

I. INTRODUCTION

Major roadblocks arise from chronic illnesses, which are characterized by prolonged effects or signs that last [**1**]. Considering that, these diseases exert an adverse impact on the lives of individuals. Human beings, firms and governmental entities are actively assisting persons affected by such illnesses [**2**, **3**]. An example of this chronic illness is hepatitis C, which is caused by an infectious agent known as hepatitis C (HCV) [**4**,**5**]. One million new instances of hepatitis C virus infection have been detected recorded this year, meanwhile According to estimates, 50 million people globally are struggling from a chronic illness. The World Health Organization (WHO) projects that cirrhosis and a form of hepatocellular carcinoma (primary liver cancer) accounted for 242,000 of the 242,000 fatalities resulted from hepatitis C in 2022. Furthermore, 58 million instances of chronic hepatitis C are expected worldwide. The majority of instances of the influenza virus are discovered in less developed and poor nations in the Middle East and Africa, in juxtaposition with wealthy nations in Europe and North America. An increased percentage of individuals suffer from chronic diseases in countries including the United Arab Emirates, China, as well as Pakistan [**6**,**7**], with Egypt reporting a significantly greater prevalence of these conditions [**8**,**9**]. Engaging increases The likelihood of contracting HCV along with other blood-borne viruses, which might be dangerous for medical staff (HCWs), because of continual exposure to microbiological dangers [**10**]. Since the widespread availability of direct-acting antiviral (DAA) drugs in 2014, HCV therapy has progressed, with an estimated efficacy incidence of 95% or higher [**11**]. The WHO claims that, by 2030, 65% fewer people will die from hepatitis and 90% fewer instances of the disease are expected. Particularly typical cause of chronic liver disease globally is **hepatitis C virus** [**12**, **13**]. Advances in (ML)[**28**] alongside (DL) in the recent past have made it feasible to diagnose HCV promptly and accurately. Techniques such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), K-Nearest Neighbors (KNN), Logistic Regression (LR), Support Vector Machines (SVMs), and Artificial Neural Networks (ANNs) lately have become widely used. By means of processing enormous datasets, these algorithms identify

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patterns that improve patient outcomes and provide more affordable options for diagnosis. Predicting therapy results and creating individualized treatment programs need the use of ML alongside DL. Robust data collection and processing is necessary since these models are effective only when substantial preprocessing, feature selection, and extraction are performed beforehand to achieve the greatest results conceivable. Hepatitis C detection has improved thanks to ML and DL, yet there are still issues with integrating classifiers on unbalanced data sets and guaranteeing reliable pretreatment. By putting in place a thorough infrastructure and assessing SVM and LSTM models, this work tackles these problems. Generating great accuracy (98.33% for SVM and 91.67% for LSTM), it offers comprehensive performance indicators along with additional information. Additionally, this study emphasizes how crucial preliminary steps include reduced dimensionality and outlier handling are to enhancing model dependability.

- **Comprehensive Data Preprocessing:** This study employed a robust data preprocessing pipeline involving outlier treatment (IQR), missing data handling, PCA, and standardization to enhance data quality and model performance.

- **Model Analysis:** By contrasting the SVM and LSTM models' performances on the automated dataset, the research investigation offers insightful information and shows how well both models excel in reaching high accuracy of 98.33% and 91.67 .

- **In-depth Performance Evaluation:** This research offers a comprehensive evaluation of model functionality with the use of accuracy, precision, recall, F1-score, confusion matrix, along with ROC curve, provide a more thorough comprehension of the model's advantages and disadvantages. Also, this study explores the potential of SVM for the given problem, contributing to the understanding of model suitability in different contexts.

This study is divided into four sections. An overview of earlier studies in the area of interest is given under the subsection on **Literature review**. The **Materials and Methods** section describes the technique used in this work, including a thorough explanation of the data. **Results and Discussion** section contains the examination of the experimental information. The **Conclusion** section, which comes last, provides a summary of the main findings along with suggestions for further study.

II. Literature Review

On the brink of producing accurate estimates utilizing the results of breakthrough study on hepatitis C prediction. The inconsistencies amid the promising prospective possibilities and prior successes are filled up by this appraisal. The research that has taken place done in a comparable fashion by other scholars is displayed below. Once the scholars validated their findings, they conducted a number of papers on the categorization of hepatitis. Ali et al.'s [**14**] plan combines fuzzy logic, decision trees (DT), and data extraction to prevent and forecast HCV outbreaks. Using the 98.1% trapezoidal fuzzy number (TFN) as opposed to DT's 92.5% prediction precision. Nandipati et al. [**15**] present an HCV predicting approach that takes into account k-nearest neighbors as well as specific random forest classifiers developed for use with the dataset accessible in the UCI ML repository [15]. They comprised 668 people in groups 0 and 1 with mild to severe liver conditions. R and Python were used to create applications with different features and attributes. Combining the components might make it practicable to create more effective immunoglobulin sequences that affect the HCV reaction [**16**]. The key components of the data were anticipated for research segments that employ RF categorization using machine learning techniques. Hashem et al. [**17**] have developed machine learning approaches for determining hepatocyte malignancy in patients with prolonged liver harm caused by HCV. A collection of processed input parameters and the CART technique (classification and regression model) are supplied. These have been arranged in order of most optimum segment of elements using LR, DT, and other techniques. For DT, and LR, the accuracy percentages were consistently 96%, and 95.5%, respectively. Das et al. sought to categorize hepatitis using velocity improved whale maximum effectiveness. Using the hepatitis dataset put on by UCI, the researchers achieved 92.5% accuracy [**18**]. A machine learning-based testing study was conducted by Abd El-Salam et al. on 4962 HCV patients in Egypt from 2006 to 2017 [**19**]. They examined 2218 individuals with gastroesophageal varices a condition that didn't happen in 2744 patients using 24 clinical testing characteristics. Their approach made use of six popular classifiers: DT, neural networks (NN), RF, NB, SVM, and Bayesian networks (BN). The national treatment initiative for those diagnosed with viral liver illness was managed by the Egyptian Ministry of Health, which made use of information gathered by the Egyptian National Council for the Combat of Viral Hepatitis. The following order of values was obtained for the precision of the SVM, RF, C4.5, multi-layer perceptron (MLP), NB, and BN: 67.8%, 66.3%, 67.2%, 65.6%, 66.7%, and 68.9%. In order to determine which patients ought to be given priority for hepatitis C virus evaluation, John Rigg et al.'s study [**20**] aims to present data in support of the use of machine learning techniques to maintain electronic medical documentation. This will ease the workload for medical personnel, make the most use of the assets at hand, and save health care expenditures. To expedite the technique's learning curve, a representative group of patients without labels was acquired; as a consequence, 16.2 million undocumented cases were included in the researcher's findings.

III. Material and Methods

The current study used a range of ML and DL methods to forecast the incidence of hepatitis C; Fig. **1** illustrates the whole procedure. IT was achieved by utilizing information which was gathered via widely available, affordably cost blood tests to identify and cure people before the illness progressed. Several ML methods ended up being used to achieve the aforementioned goal, such as SVM, NB, KNN, Logistic Regression, AdaBoost, RF, and DT, and finally, deep learning

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approaches implement strategically selected LSTM. These algorithms were chosen because of how well they could diagnose medical conditions. In contrast to earlier investigations, this study incorporates a thorough preparatory pipeline that includes normalization to improve emulate reliability, PCA for dimensionality reduction, missing data handling, along with identified outliers. Furthermore, an inverse comparison of ML and DL models is carried out, providing more in-depth understanding of their efficacy in diagnosing hepatitis C. Additionally, the research investigation places a high priority on multi-metric measurement of performance, making sure that ROC curves, accuracy, precision, recall, F1-score, confusion matrices, and F1-score are examined for a more comprehensive review.

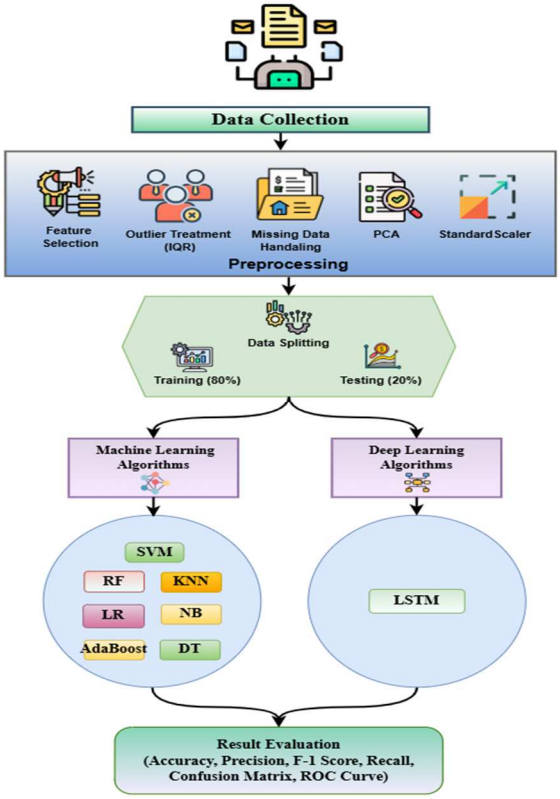


Fig. 1. Implementation of the suggested Hepatitis C Prediction System

A. Dataset

The research project collected blood samples from individuals who donated blood and individuals with hepatitis using a virtual data collection application [**21**]. In accordance with assured blood features, age, gender, and sickness standing, 615 individuals 338 females and 377 males with ages varying from 19 to 77 years old—have been determined within the gathering of data. Ishak et al.'s hepatic activity index was used to categorize patients into three groups [**22**]. Class C1: relatively mild signs of capillary fibrosis and little fibrosis in stages F0 and F1 of hepatitis Stages F2 through F6 of Class C2: Fibrosis that is crucial to treatment Class C3: Kidney-Pugh stage C end-stage liver cirrhosis which has implications pertinent to liver transplantation. 377 males and 238 females make up the entire group of 615 individuals being surveyed for the statistics. According to the data, there are 139 more men than women Fig. **2** displays the shape of the plot by sexual orientation.

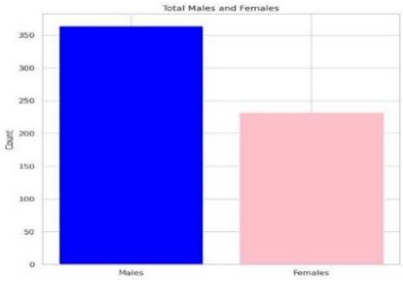


Fig. 2. Gender Distribution Bar Chart

B. Data Preprocessing and Filtering

The dataset is first imported in order to perform data processing and visualization. The dataset is subsequently analyzed to build a frame of information with a snapshot of the data topology. To prevent duplication, columns with distinct identifiers are eliminated, and rows with crucial missing values are either dropped or have their mean values imputed for designated columns. Fig. **3** illustrates how outliers in the available data are capped using the interquartile range (IQR) technique to reduce the effect they have on the reliability of the model.

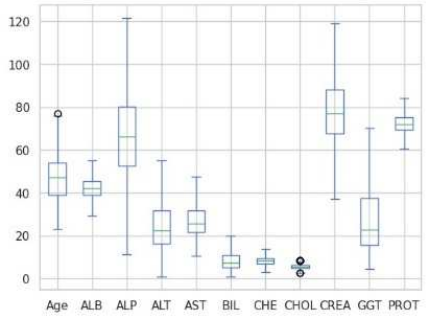


Fig. 3. Box Plot of the Dataset Before and After Outlier Capping.

Machine learning systems must work properly to be effective, the categorical variables, such "Category" and "Sex," must be transformed into quantities. Next, this information is divided between testing and training units. To ensure that the attributes' means are zero and their standard deviations are one, they are standardized. Following another decline in the volume of the data using principal component analysis (PCA), a scree plot illustrating the variance that has been
explained is displayed in fig. **4**. Thanks to this precise pretreatment before fermentation, the dataset will be certain to be transparent, organized, and ready for the subsequent phases of machine learning model training and testing.

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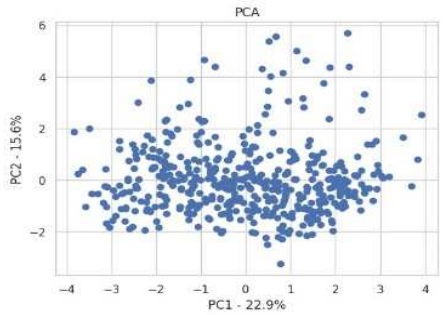


Fig. 4. Principal Component Analysis: Scree Plot and PCA Scatter Plot.

C. Machine Learning Models

Precisely part of this research, researchers used a range of machine learning techniques to predict the occurrence of hepatitis C. It was achieved by identifying and treating individuals who had the condition early on using data from routine, affordably cost blood testing. A variety of machine learning, ensemble machine learning, along with deep learning algorithms, including SVM, RF, AdaBoost, Logistic Regression, KNN, Naive Bayes, DT, and additionally LSTM were thoughtfully selected by the research team.

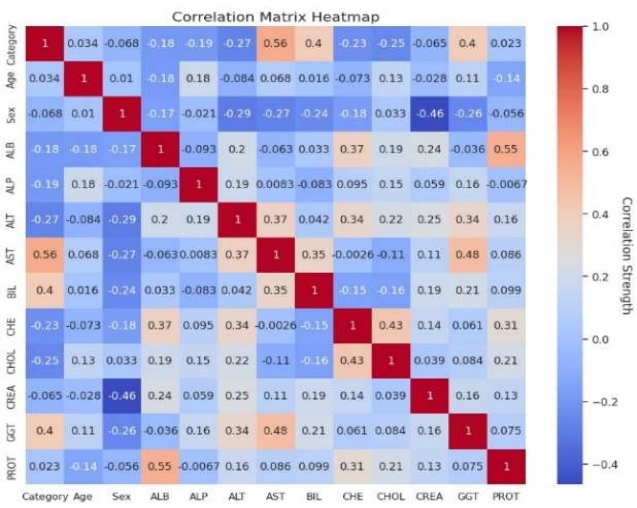


Fig. 5 Correlation Matrix with Heatmap of Dataset

SVM: One well-liked machine learning method for tasks relating to classification is the Support Vector Machine (SVM) algorithm. It operates by identifying the most effective hyperplane in a dataset for dividing various classes. Optimizing the margin between classes is the fundamental principle of Support Vector Machines (SVM) and aids in improved generalization to new data points, because SVM uses several kernel functions to manage both exponential and sequential data, it is adaptable and successful in high-dimensional domains. Finding the best hyperplane in the feature space to divide the classifications remains the technique's goal. The measurement of the circumference is computed using the assistance vectors, which are an accumulation of data segments that are closest to the hyperplane. The equations of SVM [**27**] are presented below.

$$\boldsymbol{W^T D_i + b = 0} \qquad \textbf{(1)}$$

$$\boldsymbol{Y_i . W^T D_i + b \geq 1} \qquad \textbf{(2)}$$

$$\boldsymbol{(\dot{w},\dot{D}_l) = \min \frac{\|w\|}{2} + C_i . \sum_{i=1}^{k} \lambda_i .} \qquad \textbf{(3)}$$

By optimizing the margin between two classes, the SVM decision function shown in equation **1** creates a hyperplane that divides them. Equation **2** denotes the need for positive support vectors lying on the margin border. Equation **3** shows the goal of the SVM optimization issue is to lower the weight vector's magnitude while still meeting the classification requirements.

IV. Result and Discussion

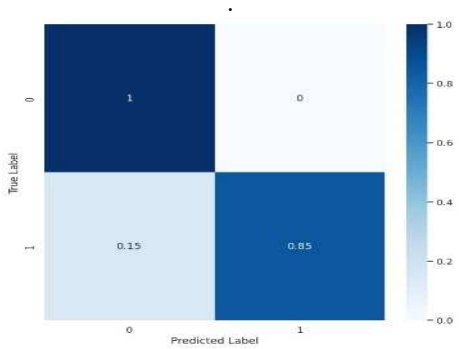
A. Result Analysis

Accuracy, precision, recall, and F1 score were estimated to feed every simulation using the confusion matrix, as demonstrated in Table **1**. This compares the effectiveness of many machine learning and deep learning methods for identifying HCV. As can be observed in Fig. **7**, higher AUC values indicate better differentiating between the positive and negative categories in the ROC curves comparing the performance of many classifiers. A brief summary of the metrics utilized by numerous deep learning and machine learning algorithms to discover the Hepatitis C virus is given in Table **1**. This table makes it easy to compare the effectiveness of each model by highlighting its accuracy, recall, precision, and F1 score. The resilience and dependability of machine learning and deep learning approaches in identifying Hepatitis C are demonstrated by the excellent metrics spanning algorithms. The SVM methodology's superior accuracy suggests that it is very good at predicting HCV, which is important for early diagnosis and treatment. This could improve medical results and slow the transmission of the virus. The quality of the dataset and the approaches' suitability for practical situations are confirmed by comparable results within algorithms. Considering an overall accuracy of 98.33, precision of 98.36, recall of 98.33, and F1-score of 98.27, the SVM methodology for classification performs quite well. With a total repeatability of 93.33, precision of 93.93, recall of 93.33, and F1-Score of 92.43, random forests likewise exhibit exceptional performance. Additional algorithms, including LR, CNN, DT, NB, and AdaBoost, respond differently, with accuracy scores ranging from 94.16 to 96.67. Additional information about the effectiveness of these models may be obtained by examining the matrices of confusion in Fig. 6. As seen in Fig. **6**, the SVM classifier produced 1true negatives, 0 false positive, 0.15 false negatives, and 0.85 true positives. With 0.99 true negatives, 0.01 false positive, 0.54 false negatives, and 0.46 true positives random forest also performed admirably.

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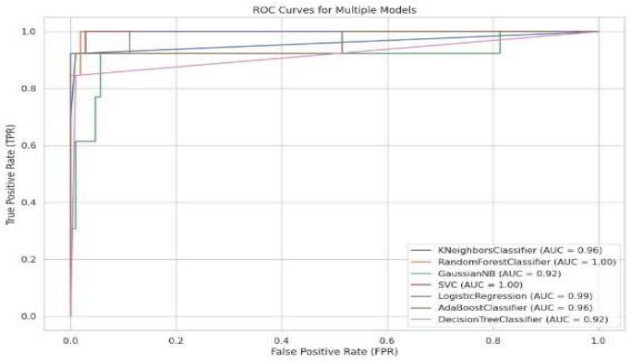
TABLE I. PERFORMANCE METRICS SUMMARY OF ML CLASSIFICATION MODELS

Models	Accuracy	Precision	Recall	F1-Score
SVM	98.33	98.36	98.33	98.27
RF	93.33	93.93	93.33	92.43
Logistic Regression	96.67	96.79	96.67	96.30
KNN	95.83	96.01	95.83	95.38
DT	94.16	93.99	94.17	94.16
Naïve Bayes	95.83	95.72	95.83	95.76
AdaBoost	94.17	94.53	94.17	99.19



SVM

Fig. 6 Confusion Matrix for SVM Algorithms



ROC Curve for 7 ML Algorithms Classifier

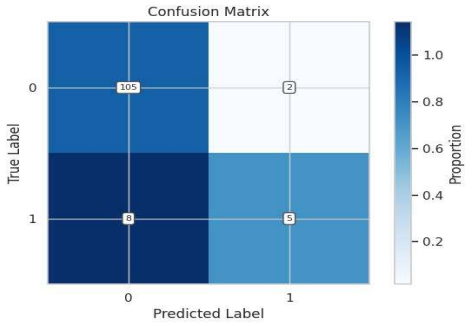
Fig. 7: Performance Evaluation: ROC Analysis of 7 different Classifier

The ROC curves in Fig**. 7** provide a comprehensive overview of the performance of many strategies. An AUC of 1.00 was obtained by the Random Forest classifiers and SVC, indicating flawless or almost perfect performance. Moreover, the Logistic Regression classifier performed admirably, with an area under the curve of 0.99. KNN and AdaBoost also performed modestly with an area under the curve of 0.96. By contrast, the AUC values of 0.92 for the GaussianNB or decision tree were lower, demonstrating a discernible decline in performance.

TABLE II. PERFORMANCE METRICS SUMMARY OF DEEP LEARNING MODEL

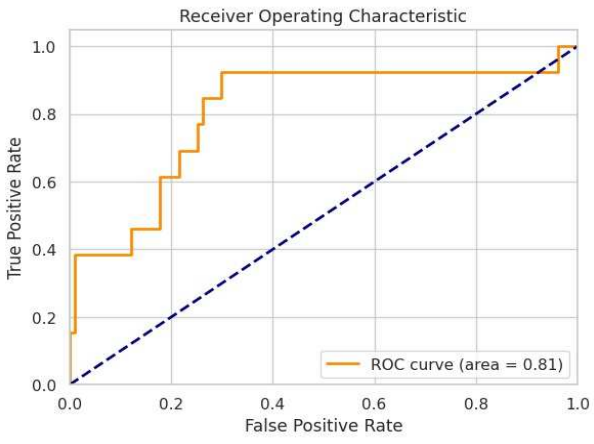
Models	Epocs	Accuracy	Precision	Recall	F1-Score
LSTM	200	91.67	71.43	38.46	50.00

Table **2** shows the Deep learning result with 1 architecture which is LSTM. LSTM shows the accuracy of 91.67%, Precision of 71.43% F1-score of 50.00% and Epocs of 200%. LSTMs Confusion matrix is shows in Fig. 8 and ROC curve also shows in Fig**. 9**. Finally, Table. **3** shows the Analyzing several ML algorithms to forecast the existence of hepatitis C with the proposed method and result. These results imply that SVM is more accurate in predicting the Hepatitis C virus, especially because of its balanced performance measures and higher precision.



LSTM

Fig. 8 Confusion Matrix for LSTM



LSTM

Fig. 9 Performance Evaluation: ROC Analysis of LSTM

TABLE III. EVALUATING SEVERAL MACHINE LEARNING ALGORITHMS TO FORECAST THE EXISTENCE OF

Author	**Dataset**	**Method**	**Accuracy**
Yusuf and Akande [**23**]	UCL	SVM, Gaussian NB, LR, DT, KNN, and MLP	87.00%
Peng et al. [**24**]	UCL	LR, DT, KNN, XGBoost, SVM and RF	91.90%
Das et al. [**19**]	UCL	Velocity-Enhanced whale Optimization	92.5%
Proposed Model	***UCL***	***SVM (ML)***	***98.33 %***

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V. CONCLUSION

This study evaluated prominent machine learning techniques for classification based on key metrics such as F1-score, precision, recall, and reliability to predict survival outcomes from medical assessments. Among the models tested, SVM achieved the highest accuracy (98.33%), while LSTM attained 91.67%. Out of all the models that were evaluated, SVM had the best accuracy (98.33%), followed by LSTM with 91.67%. SVM is the best classification technique since it also showed excellent accuracy (98.36%) and recall (98.33%). According to the results, machine learning models—specifically, SVM and LSTM—can greatly improve the identification of HCV infection, assisting public health experts and medical personnel. To increase prediction accuracy, future studies should investigate hybrid deep learning and machine learning models. Model dependability may be further increased by enlarging the dataset and adding interpretability tools such as SHAP and LIME. Furthermore, sophisticated neural network methods and real-time data integration may enhance early and accurate HCV identification.

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